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REC-CIS

$n$  lines follow with three integers on each separated by single spaces - *length*, *width* and *height*, which are length, width and height in feet of the  $i$ th box.

Constraints

$1 \leq n \leq 100$

$1 \leq \text{length}, \text{width}, \text{height} \leq 100$

Output Format

For every box from the input which has a height lesser than **41** feet, print its volume in a separate line.

Sample Input 0

```
4
5 5 5
1 2 40
10 5 41
7 2 42
```

Sample Output 0

```
125
80
```

Explanation 0

The first box is really low, only **5** feet tall, so it can pass through the tunnel and its volume is **5 x 5 x 5 = 125**.

The second box is sufficiently low, its volume is **1 x 2 x 4 = 80**.

The third box is exactly **41** feet tall, so it cannot pass. The same can be said about the fourth box.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main(){
3     int a;
4     scanf("%d",&a);
5     for(int i=0;i<a;i++){
6         int l,w,h;
7         scanf("%d %d %d",&l,&w,&h);
8         if(h<41){
9             printf("%d\n",(l*w*h));
10        }
11    }
12 }
```

|   | Input                                     | Expected  | Got       |   |
|---|-------------------------------------------|-----------|-----------|---|
| ✓ | 4<br>5 5 5<br>1 2 40<br>10 5 41<br>7 2 42 | 125<br>80 | 125<br>80 | ✓ |

Passed all tests! ✓



$$S = 0 \cdot p \cdot (p - a) \cdot (p - b) \cdot (p - c) \text{ where } p = (a + b + c) / 2$$

## Input Format

First line of each test file contains a single integer  $n$ .  $n$  lines follow with  $a_i, b_i$  and  $c_i$  on each separated by single spaces.

## Constraints

$$1 \leq n \leq 100$$

$$1 \leq a_i, b_i, c_i \leq 70$$

$$a_i + b_i > c_i, a_i + c_i > b_i \text{ and } b_i + c_i > a_i$$

## Output Format

Print exactly  $n$  lines. On each line print 3 integers separated by single spaces, which are  $a_i, b_i$  and  $c_i$  of the corresponding triangle.

## Sample Input 0

```
3
7 24 25
5 12 13
3 4 5
```

## Sample Output 0

```
3 4 5
5 12 13
7 24 25
```

## Explanation 0

The square of the first triangle is 84. The square of the second triangle is 30. The square of the third triangle is 6. So the sorted order is the reverse one.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 #include<stdlib.h>
3 #include<math.h>
4 int cmp(const void*a,const void*b){
5     return(*(int*)a-*(int*)b);
6 }
7 int main(){
8     int a;
9     scanf("%d",&a);
10
11     int b[a][3];
12     int c[a],d[a];
13     for(int i=0;i<a;i++){
14         scanf("%d %d %d",&b[i][0],&b[i][1],
15
16         int p=(b[i][0]+b[i][1]+b[i][2]);
17         c[i]=sqrt((p*(p-b[i][0]))*(p-b[i][1]
18         d[i]=c[i];
19     }
20     qsort(d,a,sizeof(int),cmp);
21     for(int i=0;i<a;i++){
22         for(int j=0;j<a;j++){
23             if(d[i]==c[j]){
24                 printf("%d %d %d\n",b[j][0],
25                 break;
26             }
27         }
28     }
29 }
```

|   | Input                            | Expected                    | Got                         |   |
|---|----------------------------------|-----------------------------|-----------------------------|---|
| ✓ | 3<br>7 24 25<br>5 12 13<br>3 4 5 | 3 4 5<br>5 12 13<br>7 24 25 | 3 4 5<br>5 12 13<br>7 24 25 | ✓ |

Passed all tests! ✓